```
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# CS AI - D
# Roll No. 2
# Subject : Machine Learning
# PRN : 12410541
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.datasets import load_diabetes
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score, confusion_matrix
```

```
df = pd.read_csv("/content/sample_data/diabetes.csv")
df
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1
...	...	...	...	...	...	...	...	...	...
763	10	101	76	48	180	32.9	0.171	63	0
764	2	122	70	27	0	36.8	0.340	27	0
765	5	121	72	23	112	26.2	0.245	30	0
766	1	126	60	0	0	30.1	0.349	47	1
767	1	93	70	31	0	30.4	0.315	23	0

768 rows × 9 columns

Next steps: New interactive sheet

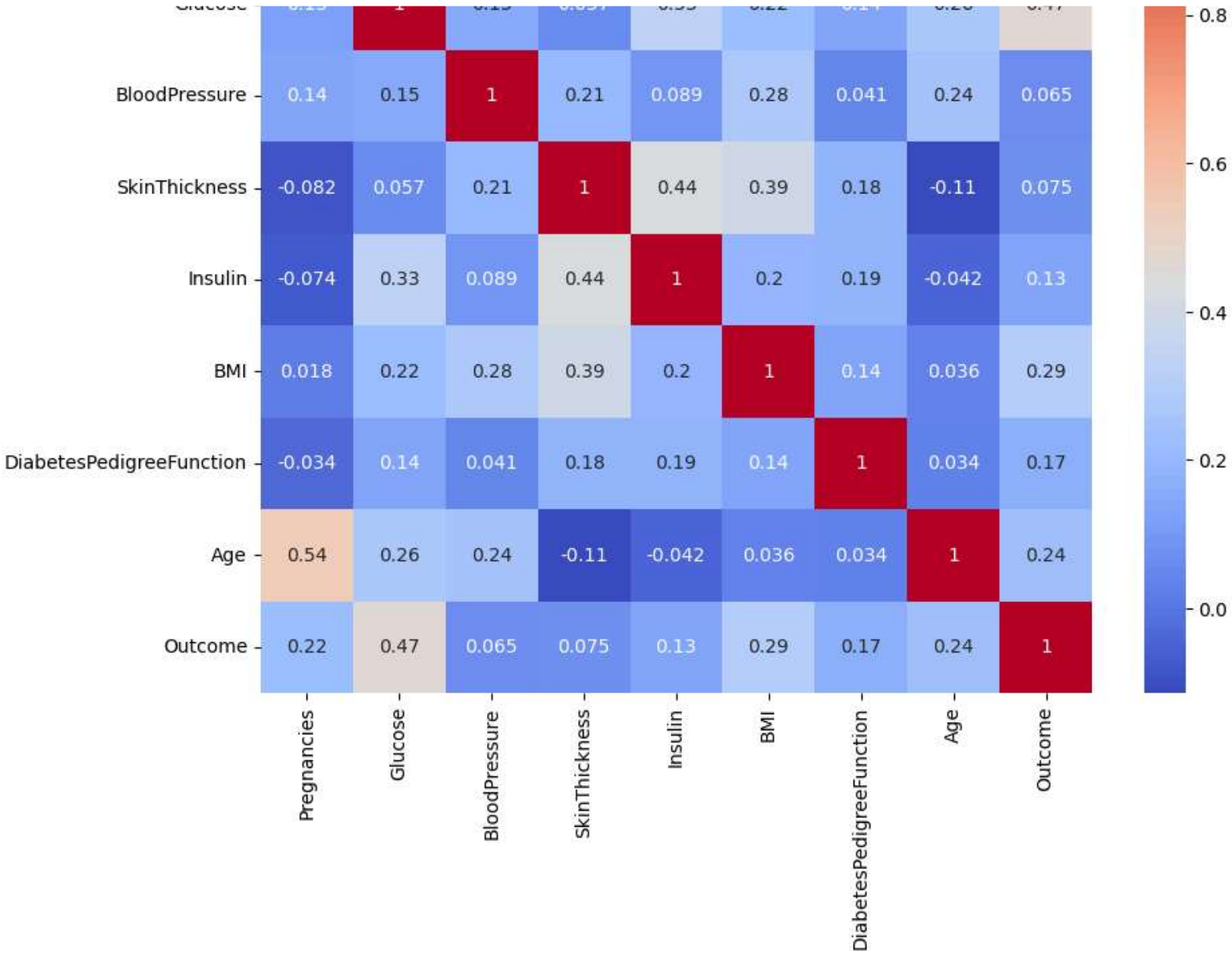
```
df.head()
```

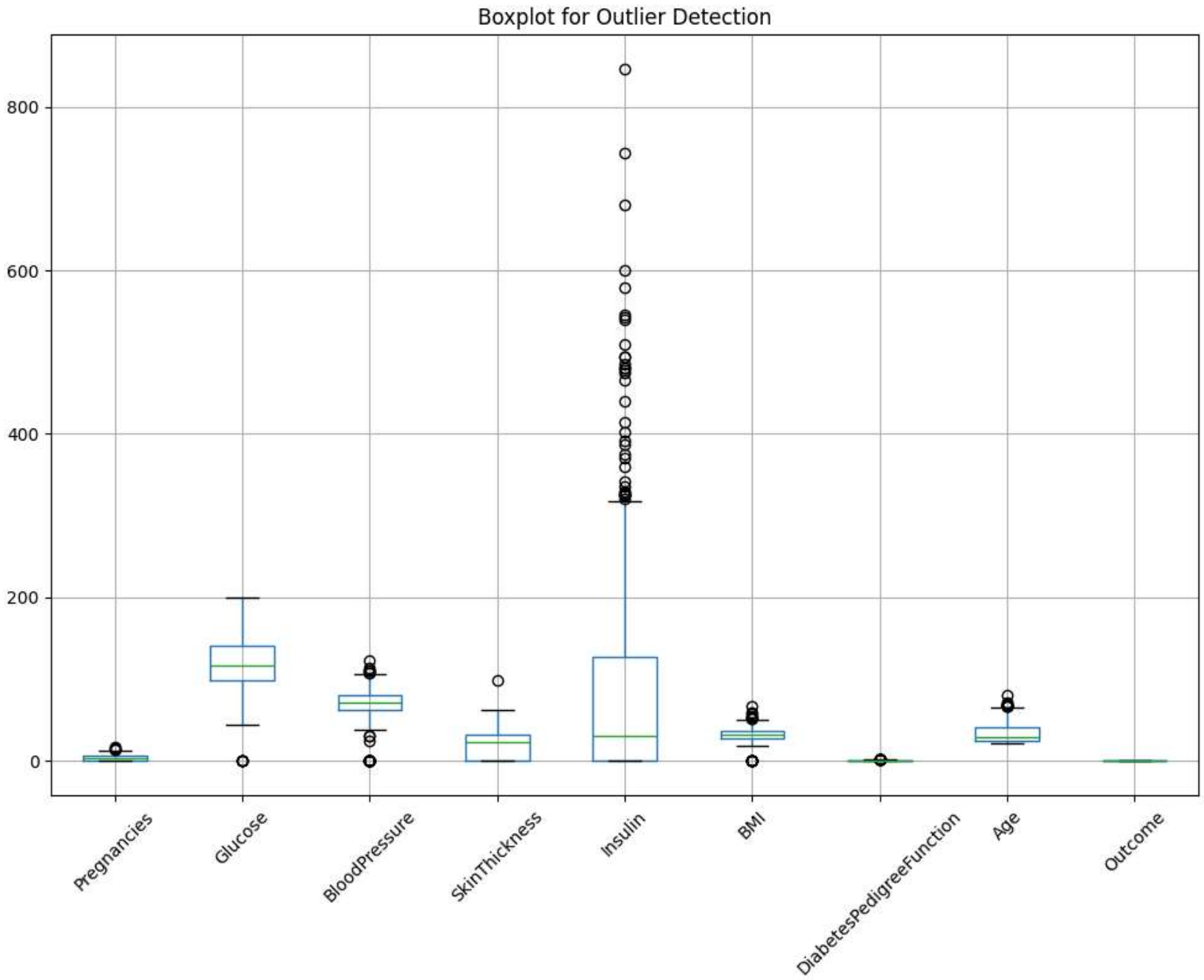
	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Next steps: New interactive sheet

```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

```
plt.figure(figsize=(12,8))
df.boxplot()
plt.xticks(rotation=45)
plt.title("Boxplot for Outlier Detection")
plt.show()
```



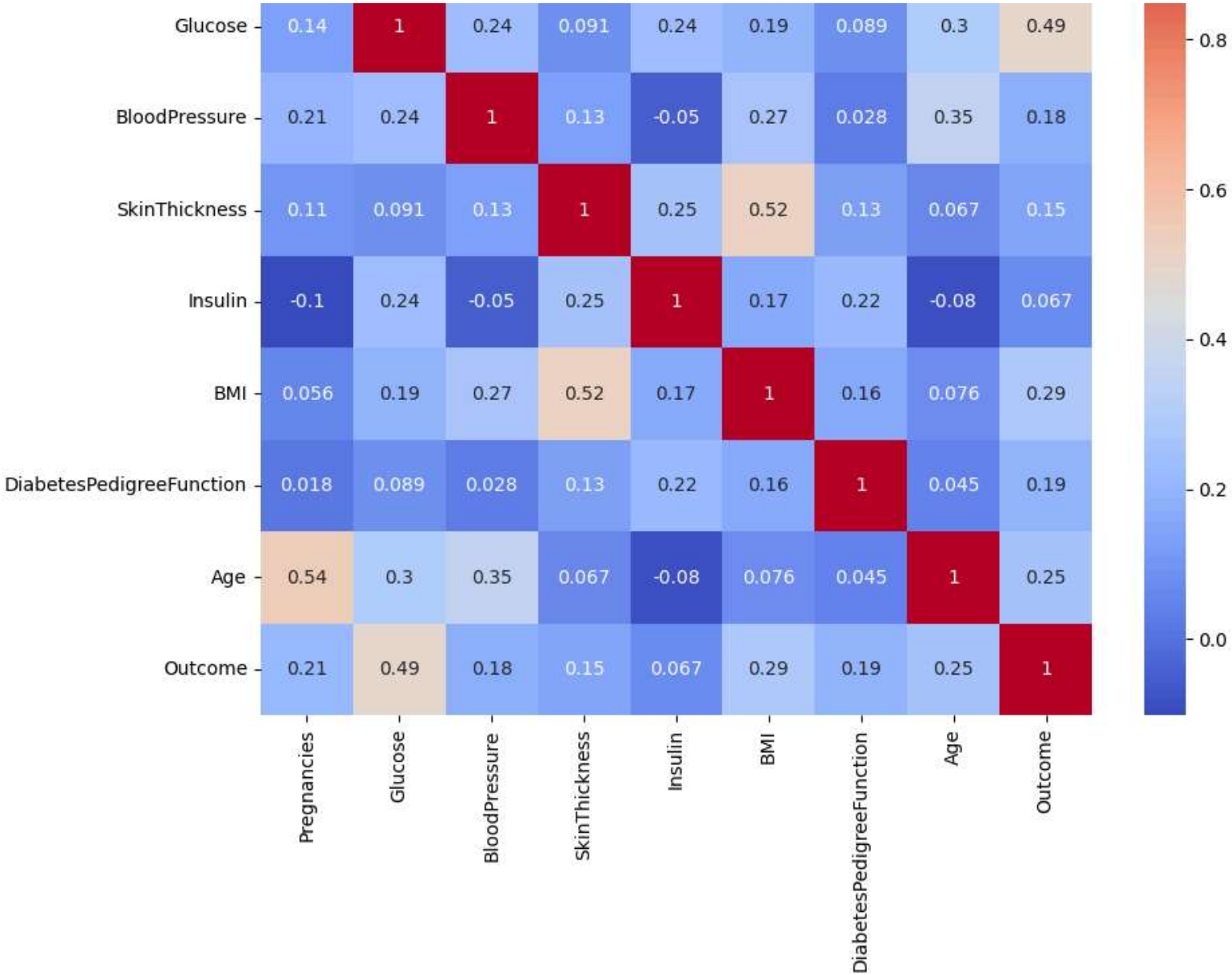

Boxplot for Outlier Detection
800
600
400
200
0
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

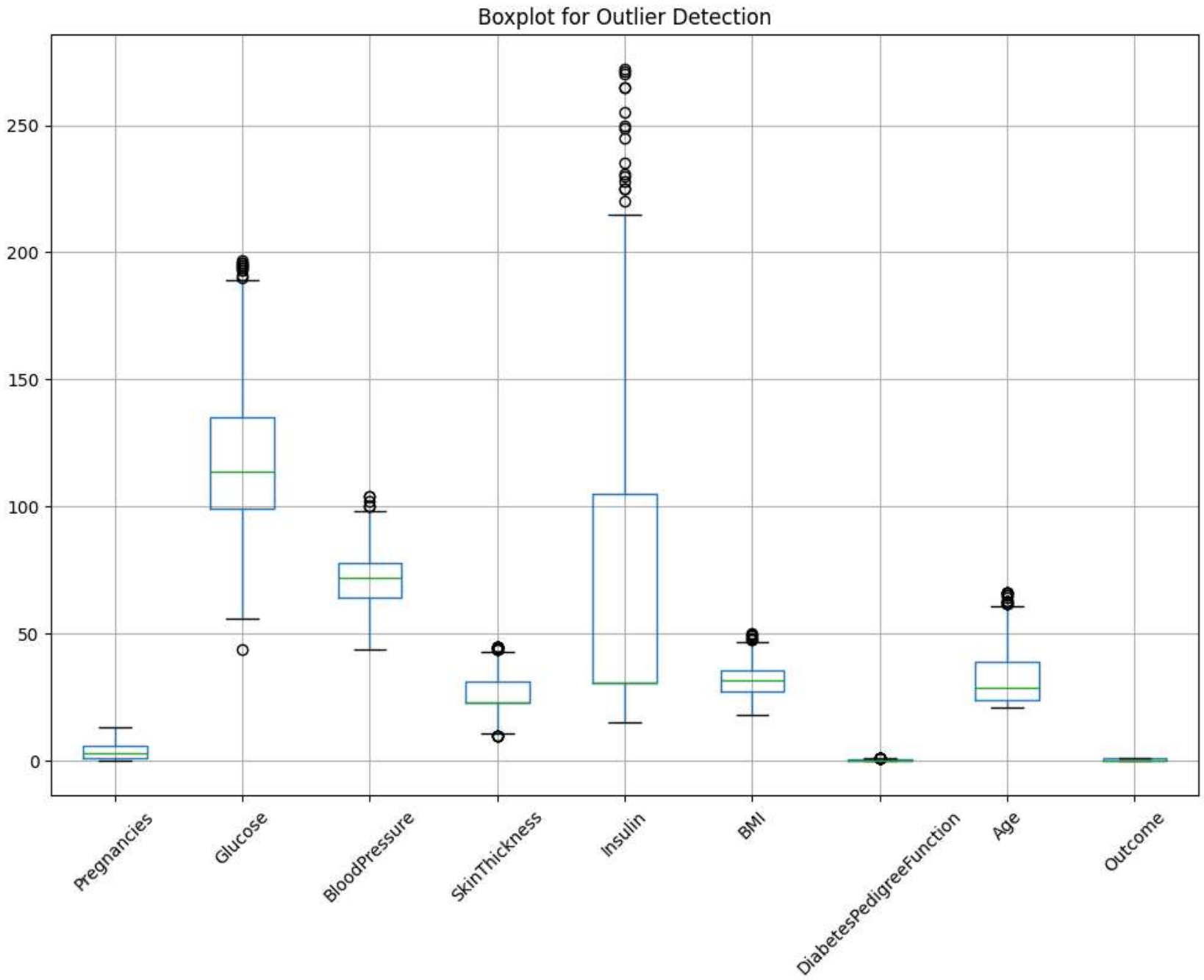
```
cols = ['Glucose','BloodPressure','SkinThickness','Insulin','BMI']

for col in cols:
    df[col] = df[col].replace(0, df[col].median())

Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1

df_dataclean = df[~((df < (Q1 - 1.5 * IQR)) | (df > (Q3 + 1.5 * IQR))).any(axis=1)]

print("Original Shape:", df.shape)
print("After Cleaning:", df_dataclean.shape)
```

```
Original Shape: (768, 9)
After Cleaning: (636, 9)
```

```
plt.figure(figsize=(10,8))
sns.heatmap(df_dataclean.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

```
plt.figure(figsize=(12,8))
df_dataclean.boxplot()
plt.xticks(rotation=45)
plt.title("Boxplot for Outlier Detection")
plt.show()
```




Boxplot for Outlier Detection
250
200
150
100
50
0
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

```
X = df_dataclean[['Glucose', 'Age']]
y = df_dataclean['Outcome']

X.head()
```

	Glucose	Age
0	148	50
1	85	31
2	183	32
3	89	21
5	116	30

Next steps: New interactive sheet

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

print("Training shape:", X_train.shape)
print("Testing shape:", X_test.shape)
```

```
Training shape: (508, 2)
Testing shape: (128, 2)
```

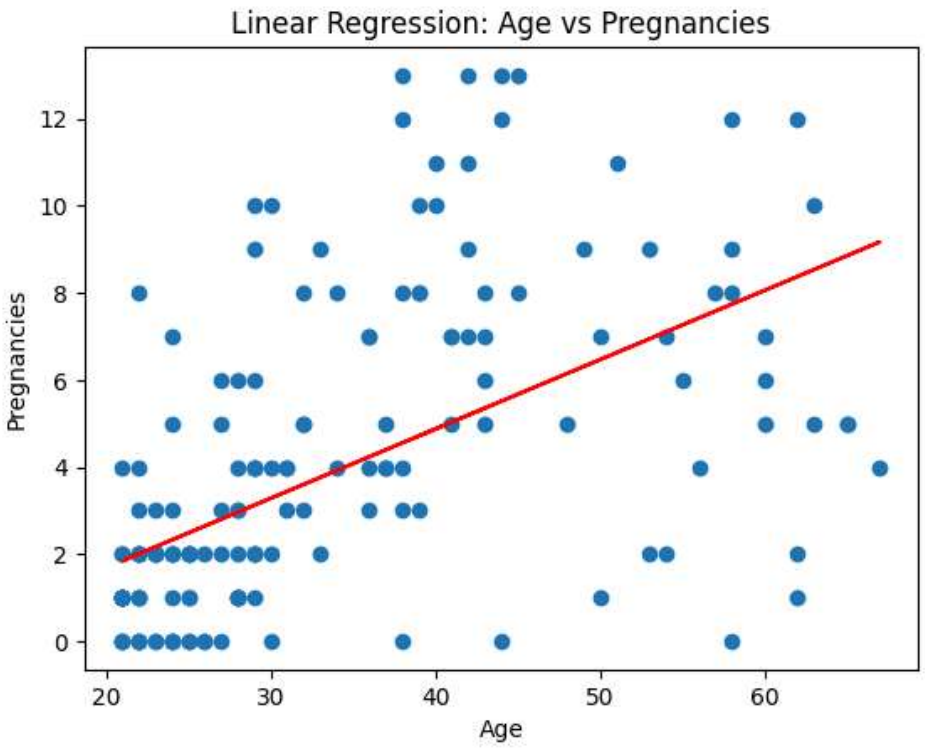
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

```
X = df[['Age']]
y = df['Pregnancies']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
```

```
MSE: 9.481139019548676
R2: 0.25012691720514646
```

```
import matplotlib.pyplot as plt

plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred, color='red')
plt.xlabel("Age")
plt.ylabel("Pregnancies")
plt.title("Linear Regression: Age vs Pregnancies")
plt.show()
```




```
from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error (MSE):", mse)
print("R² Score:", r2)
```

```
Mean Squared Error (MSE): 0.17195963836368772
R² Score: 0.22916916143620802
```

```
X = df_dataclean[['Pregnancies', 'Age']]
y = df_dataclean['Outcome']

X.head()
```

	Pregnancies	Age
0	6	50
1	1	31
2	8	32
3	1	21
5	5	30

Next steps: New interactive sheet

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

print("Training shape:", X_train.shape)
print("Testing shape:", X_test.shape)
```

```
Training shape: (508, 2)
Testing shape: (128, 2)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
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from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error (MSE):", mse)
print("R² Score:", r2)
```

```
Mean Squared Error (MSE): 0.20242559837032406
```

```
X = df_dataclean[['Age']]
y = df_dataclean['Pregnancies']
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
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from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score

print("MSE:", mean_squared_error(y_test, y_pred))
print("R²:", r2_score(y_test, y_pred))
```

```
MSE: 7.21440767033732
R²: 0.2124090960706917
```

```
X = df_dataclean[['SkinThickness', 'BMI']]
y = df_dataclean['Outcome']
```